In Silico Characterization of Prophage Regions in Publicly Available *Escherichia coli* Genomes: Exploring their Potential Relevance to Antimicrobial Resistance.

Author: Senabulya Muhammad
Affiliation: Makerere University Kampala, Uganda
Primary Track: Genomics and Bioinformatics

Background

Bacteriophages are a type of virus that infect and replicate within bacteria. This attack may leave part of the virus's genome incorporated into the bacterial genome as a prophage. The prophages play a big role in development of antimicrobial resistance in involved bacteria and their evolution as well. Characterizing prophage regions in bacterial genomes can help investigate their potential relevance to antimicrobial resistance.

Methodology

Four complete reference genomes *Escherichia coli* K12 MG1655, CFT073, O157:H7 Sakai and UTI89 were analyzed in silico. Bioinformatics command-line tool Phigaro was used to predict prophage regions, while ABRicate with the NCBI AMR-associated database and ResFinder were used for antimicrobial resistance screening. Genomic coordinates were compared to determine whether detected AMR associated matches overlapped predicted prophage regions.

Results

Phigaro predicted 28 prophage regions spanning 1,072,327 base pairs across the four genomes. O157:H7 Sakai contained the highest number of predicted regions (14; 579,247bp), whereas CFT073 contained the fewest (3; 189,422bp). ABRicate detected one antimicrobial resistance associated match in genome, while ResFinder detected no matches under the settings used. None of the four ABRicate matches overlapped the predicted prophage regions.

Conclusion

Predicted prophage abundance and genomic coverage varied among the selected *Escherichia coli* genomes. Although antimicrobial resistance associated matches were detected, none overlapped the predicted prophage regions. This provides a preliminary comparative assessment and broader genomic sampling and further validation are needed for a comprehensive study of prophage associated antimicrobial resistance.

Key words

Bacteriophages, *Escherichia coli,* Prophages, Bioinformatics, Antimicrobial resistance.